

Formula Chart Statistics

Statistics and Data Management

2025-2026

Probability

A and B mutually exclusive

$$P(A \cap B) = 0$$

Union: A or B

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Intersection: A and B

Independent

$$P(A \cap B) = P(A)P(B)$$

Dependent

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

Complement: not A

$$P(A^c) = 1 - P(A)$$

Conditional: A given B

Independent

$$P(A|B) = P(A)$$

Dependent

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Bayes' Rule: given $B_1, \dots, B_k$
(mutually exclusive and exhaustive)

$$P(B_i|A) = P(B_i \cap A)/P(A) = \frac{P(B_i)P(A|B_i)}{P(B_1)P(A|B_1) + \dots + P(B_k)P(A|B_k)}$$

Combinatorics

- **Multiplicative rule:** when drawing one element from each of k sets, with respective sizes $n_1, \dots, n_k$, the number of possibilities is $n_1 n_2 \dots n_k$
- **Permutation rule:** when drawing n different elements from a set of size N and arranging them in order, the number of possibilities is

$$\frac{N!}{(N-n)!}$$

- **Combination rule:** when drawing n different elements from a set of size N without considering order, the number of possibilities is

$$\binom{N}{n} = \frac{N!}{n!(N-n)!}$$

Quantitative data

- Population variance: $\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$
- Sample variance: $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

Discrete Random Variables

Random variable	Interpretation	Probability distribution	Expected value μ	Variance σ^2
General discrete		Table, formula or graph for $P(x)$	$\sum_{\text{all } x} xP(x)$	$\sum_{\text{all } x} (x - \mu)^2 P(x)$
Binomial	$\begin{cases} p = \text{probability of success in a single trial} \\ q = 1 - p \\ n = \text{number of trials} \\ x = \text{number of successes} \end{cases}$	$P(x) = \binom{n}{x} p^x q^{n-x}$	np	npq
Poisson	$\begin{cases} \lambda = \text{expected number of events during} \\ \quad \text{a given unit of time/area/volume etc.} \\ x = \text{number of occurrences} \end{cases}$	$P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$	λ	λ

Continuous Random Variables

In general, for a continuous random variable with probability density function $f(x)$:

$$P(a < x < b) = \int_a^b f(x)dx$$

$$P(-\infty < x < +\infty) = \int_{-\infty}^{+\infty} f(x)dx = 1$$

$$P(x = a) = 0$$

$$\text{Expected value } \mu = E(x) = \int_{-\infty}^{+\infty} xf(x)dx$$

$$\text{Variance } \sigma^2 = E(x^2) = \int_{-\infty}^{+\infty} x^2 f(x)dx - \mu^2$$

Random variable	Probability distribution	Expected value $E(x) = \mu$	Variance σ^2
Uniform between c and d	$f(x) = \frac{1}{d-c}$	$\frac{c+d}{2}$	$\frac{(d-c)^2}{12}$
Normal	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	μ	σ^2
Standard normal	$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$	0	1

Linear combinations: if $X_1, \dots, X_n$ are random variables and $a_1, \dots, a_n$ and b real numbers, then for the linear combination $X = a_1X_1 + \dots + a_nX_n + b$,

$$E(X) = a_1E(X_1) + \dots + a_nE(X_n) + b$$

and if $X_1, \dots, X_n$ are mutually independent,

$$\text{Var}(X) = a_1^2\text{Var}(X_1) + \dots + a_n^2\text{Var}(X_n)$$

Single sample tests and confidence intervals

Qualitative data

Two outcomes (S and F), target parameter = p = proportion of S . Let $\hat{p}$ = proportion of S in the sample, $\hat{q} = 1 - \hat{p}$. We assume a large sample ($n\hat{p} \geq 15$ and $n\hat{q} \geq 15$). Note: in case of a small sample ($n\hat{p} < 15$ or $n\hat{q} < 15$): use Wilson adjusted method (not discussed).

- $(1 - \alpha)100\%$ confidence interval: $\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$
- Test statistic: $z = \frac{\hat{p} - p_0}{\sqrt{p_0 q_0 / n}}$

Quantitative data

Target parameter = μ = population mean.

- Large sample ($n \geq 30$) for any population distribution:
 - ★ $(1 - \alpha)100\%$ confidence interval: $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
 - ★ Test statistic: $z = \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma}$
 - ★ Use s as estimate of σ if σ unknown
- Small sample ($n < 30$) for normally distributed population with known σ : use large-sample method.
- Small sample ($n < 30$) for normally distributed population with unknown σ :
 - ★ $(1 - \alpha)100\%$ confidence interval: $\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$
 - ★ Test statistic: $t = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s}$
 - ★ Critical value(s) calculated using a t -distribution based on $n - 1$ degrees of freedom

Two sample tests and confidence intervals

Qualitative data

Target parameter = $p_1 - p_2$. Assuming independent samples and large samples ($n_1\hat{p}_1 \geq 15$, $n_1\hat{q}_1 \geq 15$, $n_2\hat{p}_2 \geq 15$, $n_2\hat{q}_2 \geq 15$).

- Let $\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$ and $\hat{q} = 1 - \hat{p}$
- $(1 - \alpha)100\%$ confidence interval: $\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$
- Test statistic: $z = \frac{\hat{p}_1 - \hat{p}_2 - p_0}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

Quantitative data: comparing means

Target parameter = $\mu_1 - \mu_2$.

Paired samples

- Large sample ($n_d \geq 30$) for any distribution for population of differences:
 - ★ $(1 - \alpha)100\%$ confidence interval: $\bar{x}_d \pm z_{\alpha/2} \frac{\sigma_d}{\sqrt{n_d}}$
 - ★ Test statistic: $z = \frac{\sqrt{n_d}(\bar{x}_d - D_0)}{\sigma_d}$
 - ★ Use s_d as estimate of σ_d if σ_d unknown.
- Small sample ($n_d < 30$) for normally distributed population of differences with known σ_d : use large-sample method.
- Small sample ($n_d < 30$) for normally distributed population of differences with unknown σ_d :

- ★ $(1 - \alpha)100\%$ confidence interval: $\bar{x}_d \pm t_{\alpha/2} \frac{s_d}{\sqrt{n_d}}$
- ★ Test statistic: $t = \frac{\sqrt{n_d}(\bar{x}_d - D_0)}{s_d}$
- ★ Critical value(s) calculated using a t -distribution based on $n_d - 1$ degrees of freedom

Independent samples

- Large samples ($n_1 \geq 30$ and $n_2 \geq 30$) for any distributions of populations:

- ★ $(1 - \alpha)100\%$ confidence interval: $\bar{x}_1 - \bar{x}_2 \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
- ★ Test statistic: $z = \frac{\bar{x}_1 - \bar{x}_2 - D_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$
- ★ Use s_1, s_2 as estimates of σ_1, σ_2 if variances unknown.

- Small sample ($n_1 < 30$ or $n_2 < 30$) for normally distributed populations with known σ_1 and σ_2 : use large-sample method.
- Small sample ($n_1 < 30$ or $n_2 < 30$) for normally distributed populations, under the assumption that $\sigma_1 = \sigma_2$:

- ★ Let $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$
- ★ $(1 - \alpha)100\%$ confidence interval: $\bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2} \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$
- ★ Test statistic: $t = \frac{\bar{x}_1 - \bar{x}_2 - D_0}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$
- ★ Critical value(s) calculated using a t -distribution based on $n_1 + n_2 - 2$ degrees of freedom

Quantitative data: comparing variances

Target parameter = $\frac{\sigma_1^2}{\sigma_2^2}$. For independent samples under the assumption that both populations are normally distributed.

- One-sided test: without loss of generality, let $H_a : \sigma_1^2 > \sigma_2^2$
- Two-sided test: without loss of generality, let $s_1^2 > s_2^2$
- Test statistic: $F = \frac{s_1^2}{s_2^2}$
- Critical value calculated based on F -distribution with $n_1 - 1$ numerator degrees of freedom and $n_2 - 1$ denominator degrees of freedom

ANOVA

Comparing k treatment means, with

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

$$H_a : \text{at least two treatment means differ}$$

- Let
$$SST = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2, \quad MST = \frac{SST}{k - 1}$$

- Let
$$SSE = \sum_{j=1}^{n_1} (x_{1j} - \bar{x}_1)^2 + \sum_{j=1}^{n_2} (x_{2j} - \bar{x}_2)^2 + \cdots + \sum_{j=1}^{n_k} (x_{kj} - \bar{x}_k)^2, \quad MSE = \frac{SSE}{n - k}$$

- Test statistic:
$$F = \frac{MST}{MSE}$$

- Critical value for $F > F_\alpha$ calculated based on F -distribution with $k - 1$ numerator degrees of freedom and $n - k$ denominator degrees of freedom

Bonferroni multiple comparison confidence intervals

For each treatment pair i, j :
$$\bar{x}_i - \bar{x}_j \pm t_{\frac{\alpha}{2c}} s \sqrt{\frac{1}{n_i} + \frac{1}{n_j}}, \text{ with:}$$

c = number of pairwise comparisons

$$s = \sqrt{MSE}$$

$t_{\frac{\alpha}{2c}}$ calculated using a t -distribution based on $n - k$ degrees of freedom

Simple linear regression

Model and assumptions

- Model: $y = \beta_1 x + \beta_0 + \epsilon$
- ϵ normally distributed independently from x , with $E(\epsilon) = 0$ and $\text{Var}(\epsilon) = \sigma^2$

Estimators

Let

$$SS_{xx} = \sum_i (x_i - \bar{x})^2$$

$$SS_{yy} = \sum_i (y_i - \bar{y})^2$$

$$SS_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$$

- Slope β_1 is estimated as: $\hat{\beta}_1 = \frac{SS_{xy}}{SS_{xx}}$
- y -intercept β_0 is estimated as: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
- σ^2 is estimated as: $s^2 = \frac{SSE}{n-2}$, with $SSE = SS_{yy} - \hat{\beta}_1 SS_{xy}$
- Coefficient of correlation = measure for strength of the linear relationship: $r = \frac{SS_{xy}}{\sqrt{SS_{xx}SS_{yy}}}$

Tests for slope β_1

- $(1 - \alpha)100\%$ confidence interval: $\hat{\beta}_1 \pm t_{\alpha/2} \frac{s}{\sqrt{SS_{xx}}}$
- Test statistic: $t = \frac{\hat{\beta}_1 \sqrt{SS_{xx}}}{s}$
- Critical value(s) calculated using a t -distribution based on $n - 2$ degrees of freedom